

# Requirement Specification Document

## Library Management Database System

### *Week 1 – Requirement Analysis and Member Database Module*

|                        |                                    |
|------------------------|------------------------------------|
| <b>Student Name</b>    | [Your Name]                        |
| <b>Register Number</b> | [Your Register No.]                |
| <b>Project Title</b>   | Library Management Database System |
| <b>Week</b>            | Week 1                             |

## 1. Introduction

### 1.1 Purpose

The purpose of this document is to analyze and specify the business requirements of the Library Management Database System. This system is intended to manage members, books, loans, and returns for a library, ensuring accurate tracking of books from acquisition to circulation and return.

### 1.2 Project Scope

The system will allow a library to store and manage member information, book catalogs, loan (issue) details, and return/fine status. It will support core library operations such as member registration, book cataloguing, book issue, book return, and fine calculation. The system will be built using a relational database with well-defined tables and relationships.

### 1.3 Intended Users

- Members – borrow books and view their loan history
- Librarians / Administrators – manage the book catalog, issue and receive returns
- Library Staff – update book availability and shelf status

## 2. Business Requirements

The following are the key business needs the system must satisfy:

- Maintain accurate and up-to-date member records
- Maintain a catalog of books with author, category, and available copies
- Allow members to borrow (issue) one or more books at a time
- Track the status of each loan (Issued, Returned, Overdue, Lost)
- Record fine details for overdue or lost books
- Generate reports such as most borrowed books, overdue books, and active members

### 3. Functional Requirements

Functional requirements describe what the system must do.

| ID   | Requirement Description                                                                        |
|------|------------------------------------------------------------------------------------------------|
| FR-1 | The system shall allow new members to register with name, email, phone number, and address.    |
| FR-2 | The system shall allow librarians to add, update, and remove books from the catalog.           |
| FR-3 | The system shall allow members to borrow one or more books in a single transaction.            |
| FR-4 | The system shall automatically calculate the due date based on the issue date and loan period. |
| FR-5 | The system shall record fine amount and payment status for overdue or lost books.              |
| FR-6 | The system shall allow tracking and updating of loan status (Issued/Returned/Overdue/Lost).    |
| FR-7 | The system shall allow members to view their past and current borrowing history.               |
| FR-8 | The system shall allow librarians to generate a circulation report for a given date range.     |

### 4. Non-Functional Requirements

- Performance: The system should retrieve loan details within 2 seconds.
- Security: Member personal data and login credentials must be stored securely.
- Reliability: The database must maintain data consistency (e.g., available copies should not go negative).
- Usability: Data entry forms should be simple for library staff to use.
- Scalability: The database design should support an increasing number of members and books over time.

## 5. Member Database Module – Data Requirements

Since this week's focus includes the Member Database Module, the following entity and attributes have been identified:

| Attribute       | Data Type         | Description                         |
|-----------------|-------------------|-------------------------------------|
| member_id       | INT (Primary Key) | Unique identifier for each member   |
| full_name       | VARCHAR(100)      | Member's full name                  |
| email           | VARCHAR(100)      | Unique email used for login/contact |
| phone_number    | VARCHAR(15)       | Member contact number               |
| address         | VARCHAR(255)      | Residential address of the member   |
| membership_date | DATE              | Date the member account was created |

## 6. Assumptions and Constraints

- The system will be developed using a relational database (e.g., MySQL / Oracle / PostgreSQL).
- Only registered members can borrow books.
- Each loan must be linked to exactly one member.
- This document covers Week 1 requirements only; further modules (Book, Loan, Fine) will be detailed in subsequent weeks.

## 7. Conclusion

This document captures the initial business and functional requirements for the Library Management Database System, with specific focus on the Member Database Module. It will serve as the foundation for designing the ER diagram, database schema, and subsequent implementation in the following weeks.